CS 200 Project 2: ChocAn System Analysis

Fall 2026 — Due Tuesday, October 13, 2026, 11:59 p.m.
GitHub repository: https://github.com/ahsuarez-ua/CS200-ChocAn-Team-6
Team: 6
Slack Channel: #tuscaloosans-team-6

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1. Header Table

Name	CWID	Crimson Email	Hours Contributed
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Submitter	Jacob Wilson	—	—

2. Task Distribution Table

Name	Tasks Performed	% Contributed
Jacob Wilson	Task 1 analysis paragraph, main activity diagram and its description, 2 use case diagrams, and report assembly	25%
Aaron Suarez	Task 2: Stereotyped class diagram and its description	25%
Gabriel Mckinney	Task 3: Member/provider use case diagrams and Verify Provider, with descriptions	25%
Peyton Hardy	Task 3: Manager/operator use case diagrams, plus Verify Operator and Verify Manager, with descriptions	25%
Total		**100%**

Assumptions and Clarifications (Jacob Wilson)

- Verify Operator, Verify Manager, and Verify Provider were added as new use cases. Operators, managers, and providers must be verified before they can use their terminals.
- The weekly accounting procedure runs automatically every Friday at midnight, and a manager can also run Generate Reports on request.
- ACME Accounting Services handles all fee collection and member suspensions. ChocAn only reads member status, which is either valid or suspended.
- ChocAn only writes the EFT data to a file. The bank carries out the actual funds transfer.
- Provider, operator, and manager terminals are simulated in software, and swiping a card is simulated by typing in the member number.
- Use case names are the same as in Project 1 unless we revise some for this assignment.
- ACME Accounting Services is modeled as an actor in the class diagram because it updates member status through the ACMEInterface. Members do not interact with the system directly, so they appear only as an entity.

Task 1: Analysis (Jacob Wilson)

The ChocAn Data Center handles member and provider verification, billing, and weekly reports, so nothing is done manually anymore. The Data Center stores all information about members, providers, services, and billing in one central database. Concerning hardware, the system needs the Data Center server, provider terminals with a magnetic card reader and keypad, and terminals for operators and managers. The system integrates with ACME Accounting Services, which collects membership fees and updates member status, and sends a weekly EFT file to the bank. Providers swipe member cards and enter services, operators maintain records, and managers view reports.

Task 2: Stereotyped Class Diagram (Aaron Suarez)

The stereotyped class diagram shows each class's associations with other classes in the system. Actors only interact with the system through a boundary class. Boundary classes pass requests to control classes, and the control classes read and update the entity classes. The entity classes are where the data is stored. This system has five actors, five boundary classes, eight controller classes, and eight entity classes.

In this diagram, the boundary classes are ProviderTerminal, OperatorTerminal, ManagerTerminal, SchedulerUI, and ACMEInterface. These are connected to the Provider, Operator, Manager, Scheduler, and ACME Accounting Services actors respectively.

The OperatorTerminal is connected to the ProviderController because operators control the addition, deletion, and editing of providers. The ProviderController is connected to the Provider and ProviderDirectory entities because the ProviderController is what the operator uses to make the edits to the Provider and ProviderDirectory classes. The OperatorTerminal also connects to the MemberController because operators maintain member records as well. The MemberController handles the Member entity and is used by two other boundary classes, ACMEInterface and the Provider Terminal. It connects to the ACMEInterface because ACME Accounting Services uses it to suspend and reinstate members.

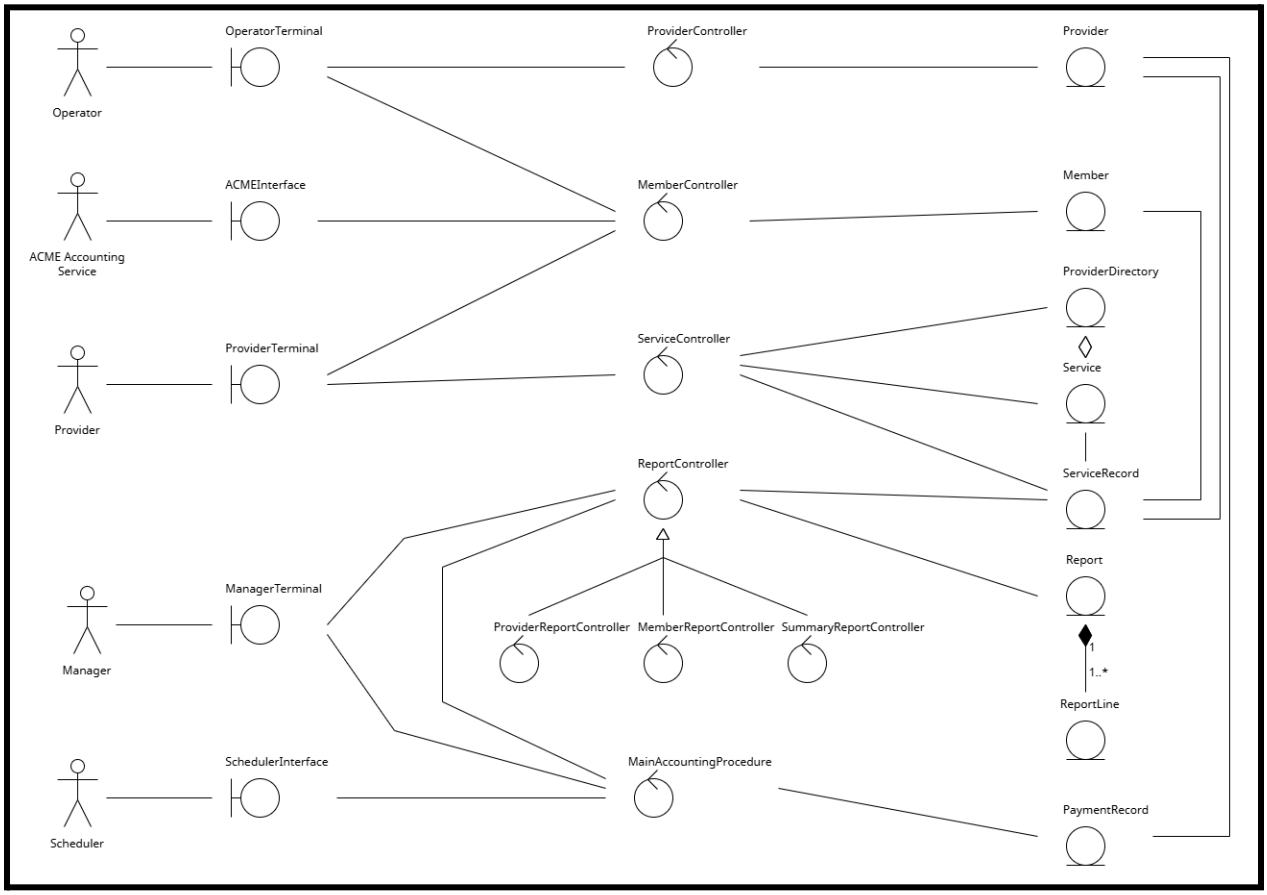
The ProviderTerminal also connects to the ServiceController. The ServiceController generates a ProviderDirectory when a Provider asks for it, writes a ServiceRecord, and handles billing.

The Scheduler interacts with the SchedulerUI to check what time that the MainAccountingProcedure is set to run, and runs it if it is that time. Normally this is set at midnight on Friday.

The Manager actor interacts with the ManagerTerminal which can run the MainAccountingProcedure when needed, and also generate reports when the Manager wants The MainAccountingProcedure interacts with the ServiceRecord and PaymentRecord to generate reports. This requires the MainAccountingProcedure to call on the ReportController to get the per-provider totals to create the PaymentRecord.

The ReportController is a superclass that the ProviderReportController, ProviderReportController, and MemberReportController inherit from. The ReportController interacts with the ServiceRecord entity class as it needs to use the data to generate each report. The Report entity class consists of a bunch of ReportLine entities. The ReportLine only exists in its report and is deleted with it. This is shown via composition on the diagram.

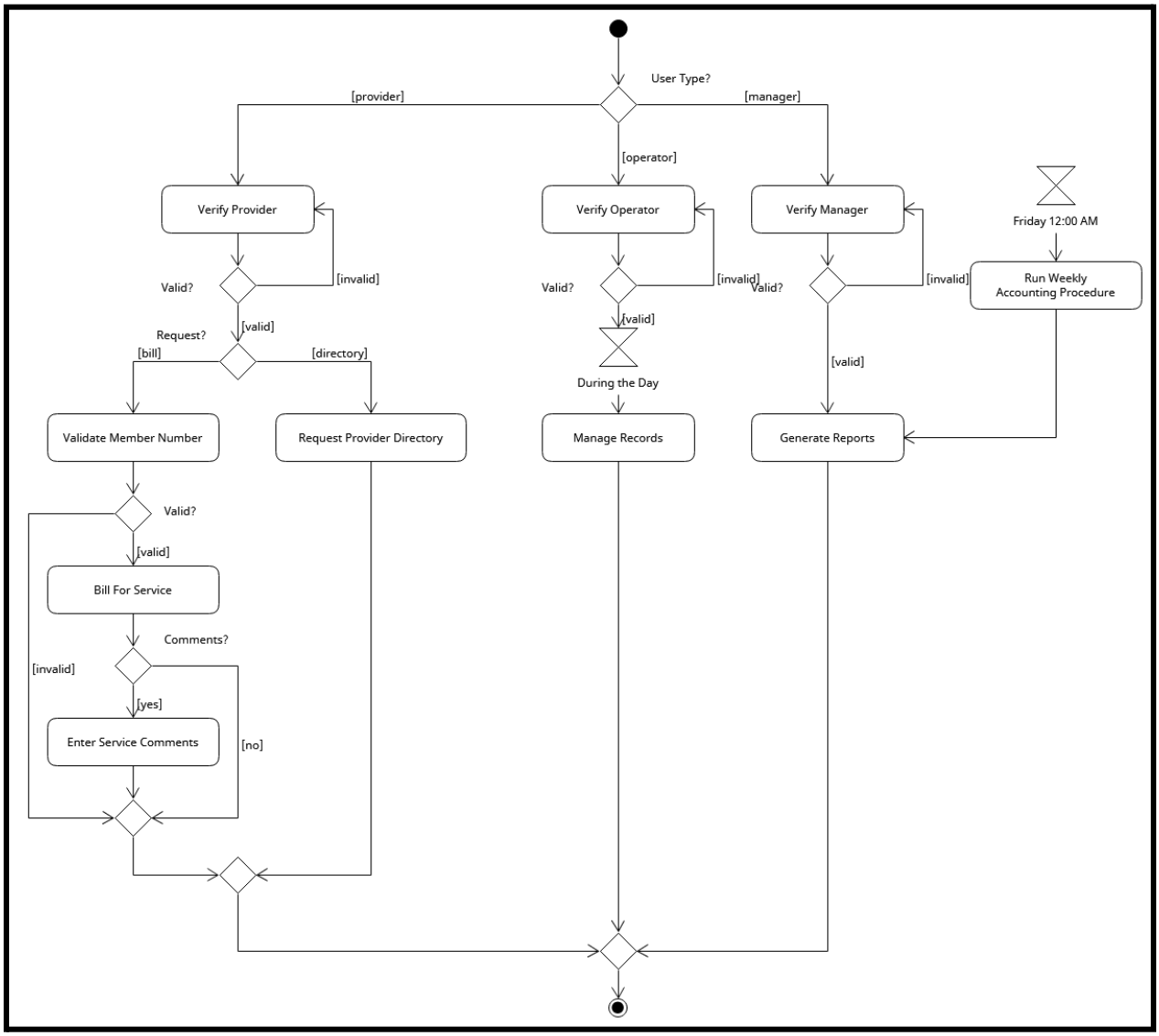
Several of the entity classes are connected to each other. Each ServiceRecord needs data from the Service, Provider, and Member entity classes. The ProviderDirectory is just a list of services, codes, and fees. This is shown as an aggregation on the diagram.



Task 3: Activity Diagrams

3.1 Main Activity Diagram (Jacob Wilson)

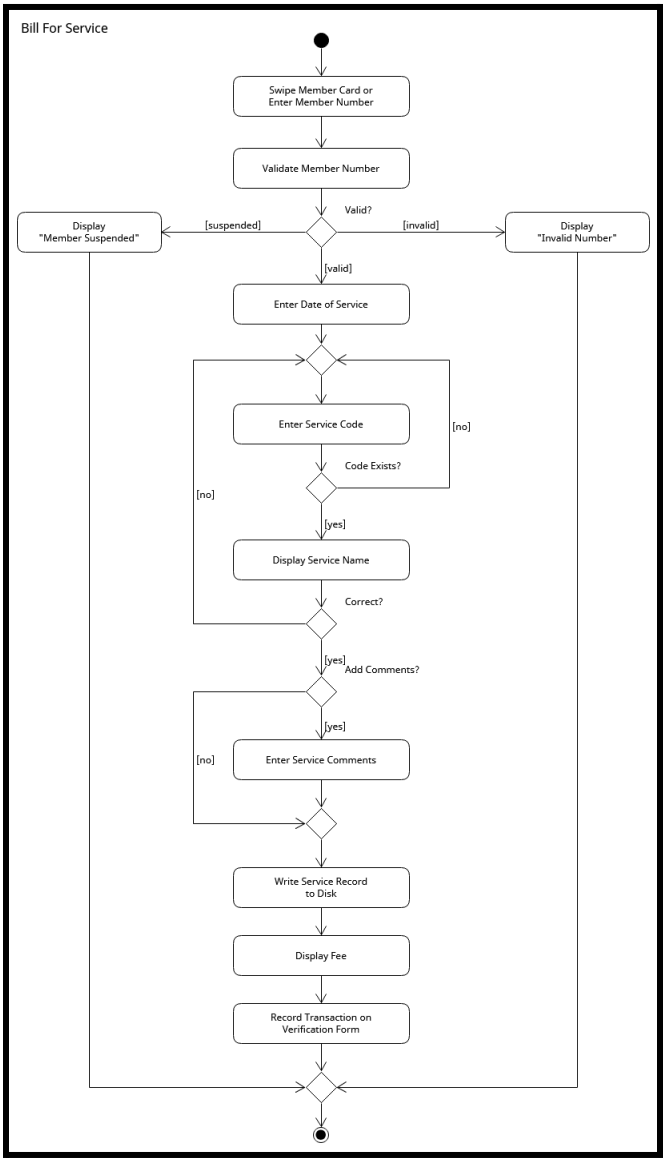
The main activity diagram serves as the starting point for all use case activity diagrams. Depending on the user type, the system directs a provider, operator, or manager to Verify Provider, Verify Operator, or Verify Manager. If verification fails, the user returns to try again. Once verified, a provider can either use Bill For Service or Request Provider Directory. For billing, the process first runs Validate Member Number, then Bill For Service, and optionally Enter Service Comments. A verified operator goes to Manage Records once the During the Day timer allows interactive mode, while a verified manager goes to Generate Reports. At 12:00 AM every Friday, a timer starts Run Weekly Accounting Procedure, which also runs Generate Reports. All process paths eventually lead to a single exit point.



3.2 Member/Provider Use Cases (Gabriel Mckinney)

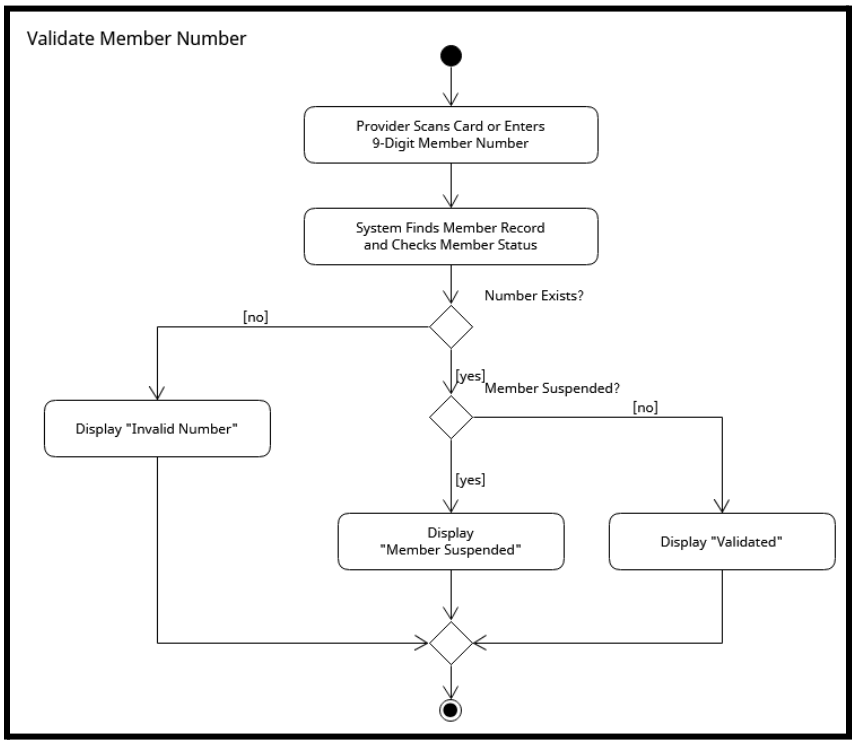
3.2.1 Bill For Service

The Bill For Service activity diagram shows how a provider bills ChocAn for a service given to a member. The provider swipes the member card or enters the member number, and the system runs Validate Member Number. If the number is invalid or the member is suspended, the system displays “Invalid Number” or “Member Suspended” and the activity ends. If the member is valid, the provider enters the date of service and the service code. If the code does not exist, the provider is returned to enter it again. The system displays the service name, and if the provider says it is wrong, they re-enter the code. The provider can optionally run Enter Service Comments. The system then writes the service record to disk and displays the fee, and the provider records the transaction on the verification form. All paths merge into a single exit node.



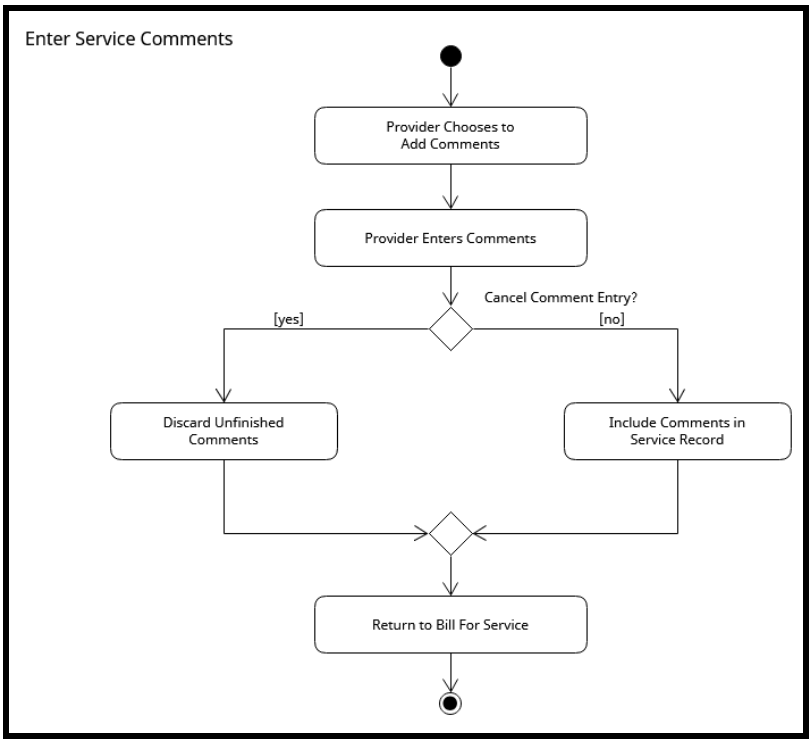
3.2.2 Validate Member Number

The Validate Member Number activity diagram shows how the system checks a member's number and status before a provider bills for a service. The provider scans the member card or enters the nine-digit member number. The system searches for the member record and checks the member's status. If the number does not exist, the system displays "Invalid number." If the member is suspended, the system displays "Member suspended." If the number exists and the member is not suspended, the system displays "Validated." All three outcomes lead to a single activity final node.



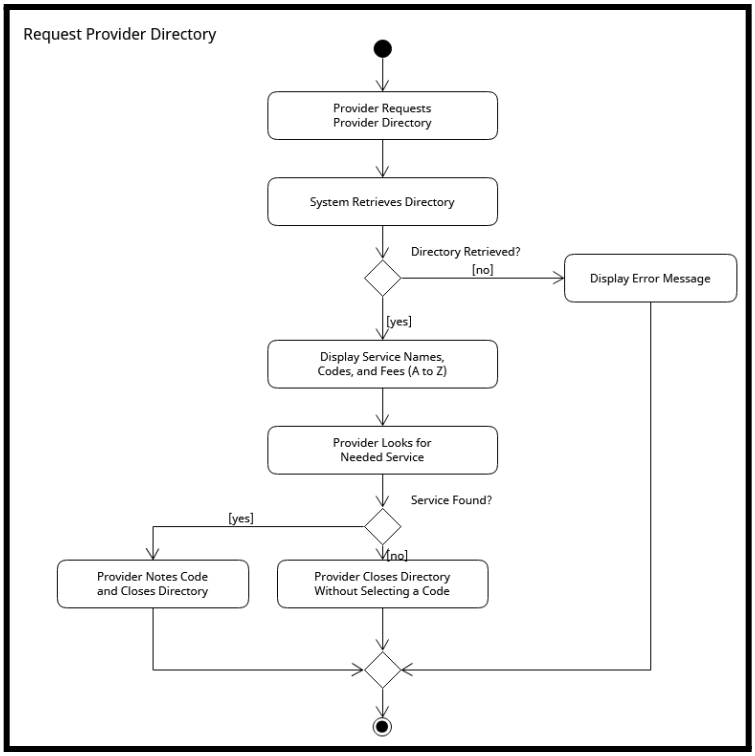
3.2.3 Enter Service Comments

The Enter Service Comments activity diagram shows how a provider optionally adds comments to a service record while billing ChocAn. After confirming the service name, the provider chooses to add comments and enters them. If the provider cancels the comment entry, the system discards the unfinished comments and returns to Bill For Service without comments. Otherwise, the system includes the comments in the service record and returns to Bill For Service so billing can continue. Both paths lead to a single activity final node.



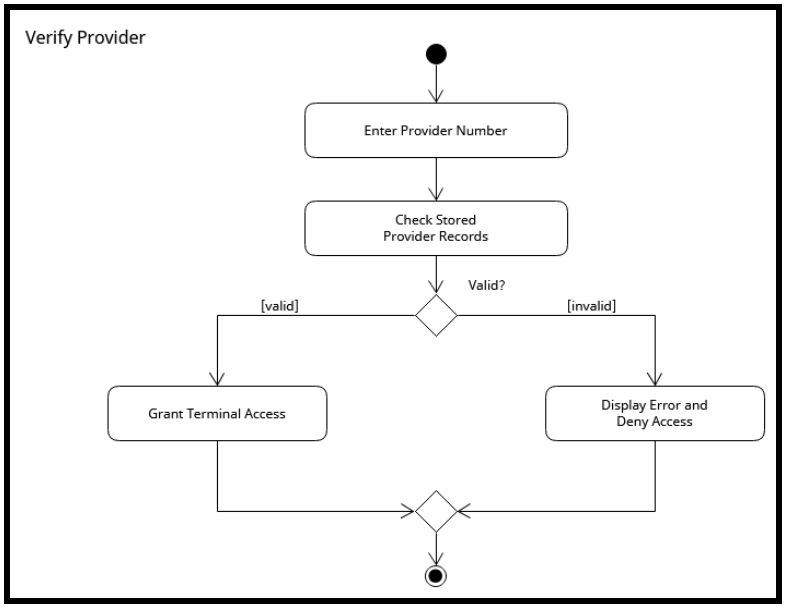
3.2.4 Request Provider Directory

The Request Provider Directory activity diagram shows how a provider accesses the directory to find a service code and fee. The provider requests the directory from the terminal, and the system retrieves it. If retrieval fails, the system displays an error message and the activity ends. If retrieval succeeds, the system displays the service names in alphabetical order with their corresponding six-digit service codes and fees. The provider looks for the needed service and notes its code if found. If the provider cannot find the service, they close the directory without selecting a code. Both successful retrieval paths and the error path lead to a single activity final node.



3.2.5 Verify Provider

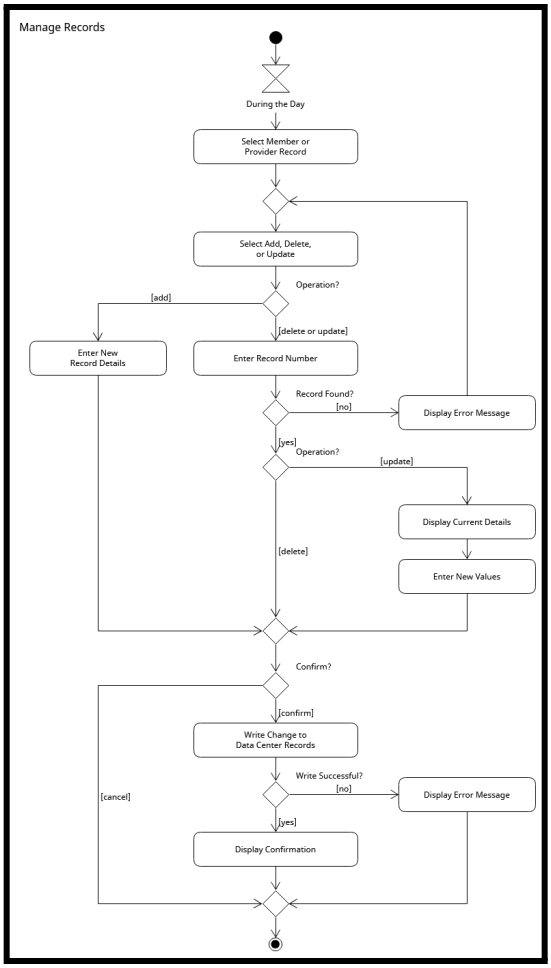
The Verify Provider activity diagram shows how the system verifies a provider before granting access to the provider terminal. The provider enters their provider number, and the system checks it against the stored provider records. If the number is valid, the system grants access to the provider terminal. If it is invalid, the system displays an error message and denies access. Both paths lead to a single activity final node.



3.3 Manager/Operator Use Cases (Peyton Hardy)

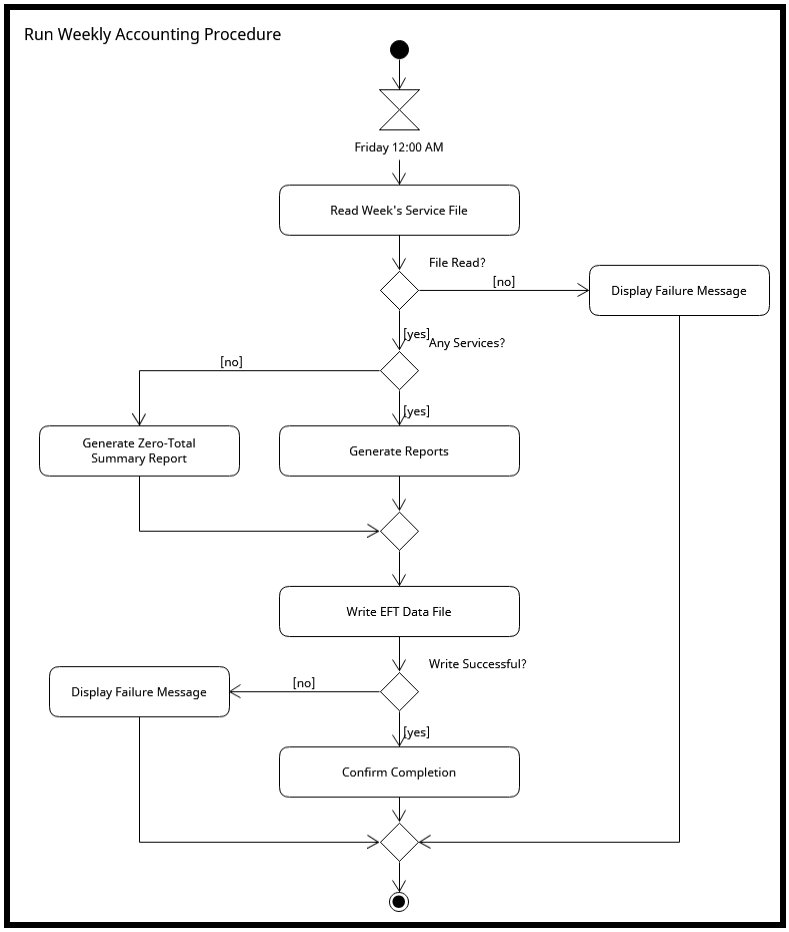
3.3.1 Manage Records

The activity diagram for Manage Records illustrates the way in which an operator can add, delete, or update member and provider records at the ChocAn Data Center. Because records can only be changed in interactive mode, the activity begins with a During the Day timer. The operator first selects a member or provider record and then picks an operation. When adding a record, the operator inputs the details of the new record. In the case of deletion or updating, the operator keys in the record number. If the system does not find a matching record, an error message is shown and the operator is asked to select an operation once more. For an update, the system shows the current details and the operator then enters the new values. The operator then has the option of confirming or canceling the change, and if they cancel, the activity ends without the change being saved. After the change has been confirmed, the system writes the change to the Data Center records. An error is displayed if the write fails, or a confirmation message is shown if it succeeds. All the different paths lead to a single exit node.



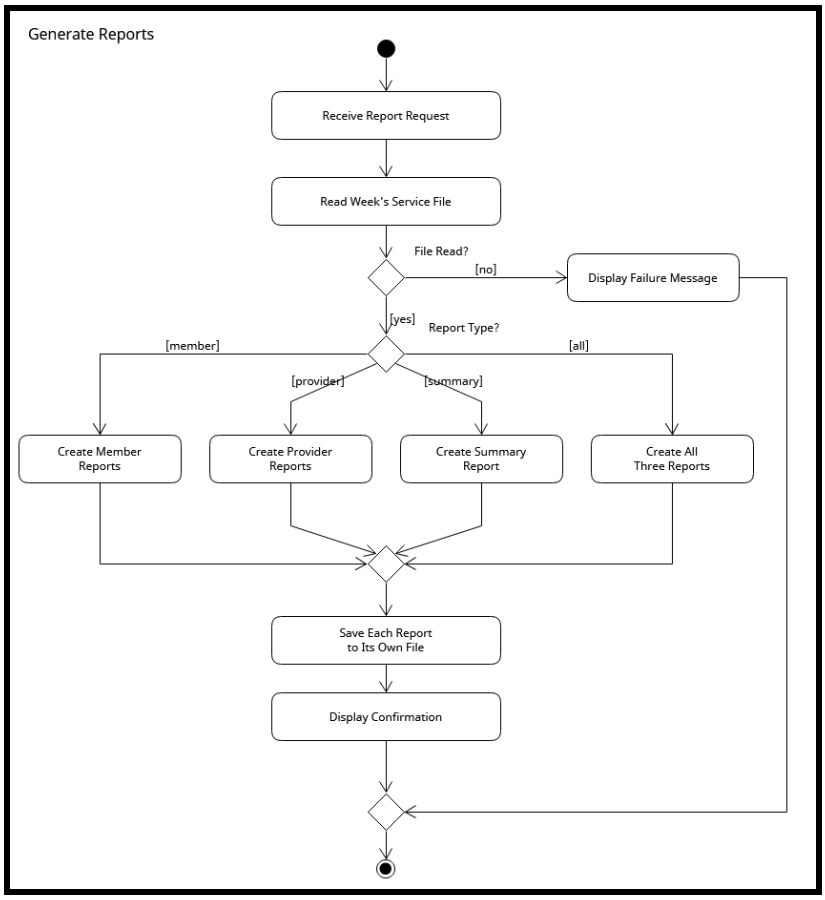
3.3.2 Run Weekly Accounting Procedure

The Run Weekly Accounting Procedure activity diagram shows the weekly accounting run at the ChocAn Data Center. The Friday 12:00 AM timer starts the procedure, and the system reads the week's service file. If the file cannot be read, the system displays a failure message and the activity ends. If there were no services that week, the system generates a summary report with zero totals. Otherwise, it runs Generate Reports. The system then writes the EFT data file. If the write fails, it displays a failure message; if it succeeds, it confirms completion. All paths merge into a single exit node.



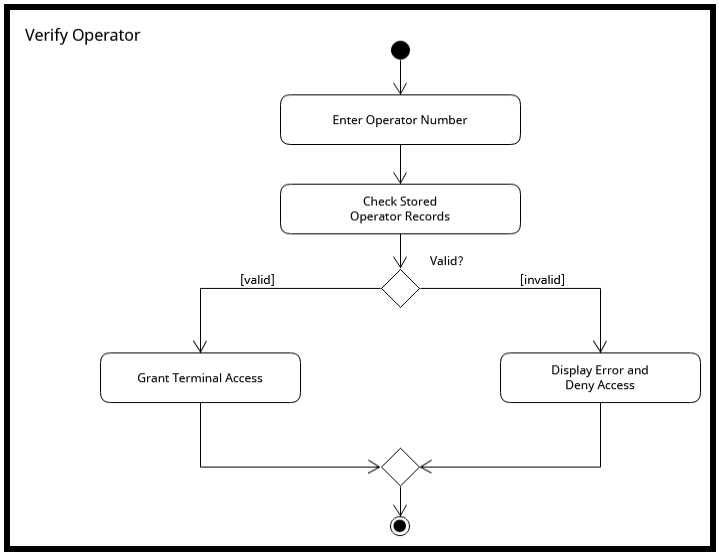
3.3.3 Generate Reports

The Generate Reports activity diagram shows how the system produces reports for a verified manager or for the weekly accounting procedure. After receiving the request, the system reads the week's service file. If the file cannot be read, it displays a failure message and the activity ends. Otherwise, the system creates member reports, provider reports, the summary report, or all three, depending on the request. Each report is saved to its own file, and the system displays a confirmation. All paths merge into a single exit node.



3.3.4 Verify Operator

The Verify Operator activity diagram shows how the system checks an operator before allowing access to the operator terminal. The operator enters their operator number, and the system checks it against the stored operator records. If the number is valid, the system grants terminal access. If it is invalid, the system displays an error and denies access. Both paths merge into a single exit node.



3.3.5 Verify Manager

The Verify Manager activity diagram shows how the system checks a manager before allowing access to the manager terminal. The manager enters their manager number, and the system checks it against the stored manager records. If the number is valid, the system grants terminal access. If it is invalid, the system displays an error and denies access. Both paths merge into a single exit node.

